

Symmetric tensor categories

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Lecture 1

The model example: $\text{Rep}_{\mathbb{k}}(G)$

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We would now like to axiomatize the categorical structures visible in $\text{Rep}_{\mathbb{k}}(G)$ without referring to underlying vector spaces.

Monoidal category: data

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Monoidal category: pentagon axiom

First, the associativity constraint is required to satisfy the **pentagon axiom**: for all $W, X, Y, Z \in \mathcal{C}$, the diagram

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commutes. Equivalently,

$$a_{W, X, Y \otimes Z} a_{W \otimes X, Y, Z} = (\text{id}_W \otimes a_{X, Y, Z}) a_{W, X \otimes Y, Z} (a_{W, X, Y} \otimes \text{id}_Z).$$

Monoidal category: unit axiom

Next, the unit object $\mathbf{1}$ is required to satisfy

$$\mathbf{1} \otimes \mathbf{1} \cong \mathbf{1},$$

and tensor multiplication by $\mathbf{1}$ on either side must be an equivalence:

$$L_{\mathbf{1}} : \mathcal{C} \rightarrow \mathcal{C}, \quad X \mapsto \mathbf{1} \otimes X,$$

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The **MacLane's coherence theorem** implies that we may ignore parentheses and unit objects inside tensor products.

Braided monoidal category: data

A **braided monoidal category** is a monoidal category $\mathcal{C}$ together with a natural isomorphism

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They imply that for every object $X \in \mathcal{C}$ the object $X^{\otimes n}$ carries a natural action of the **braid group** B_n , with generator b_i acting by c in the i -th and $i+1$ -th factors.

Braided monoidal category: hexagon axioms

1. For all $X, Y, Z \in \mathcal{C}$, the diagram

$$\begin{array}{ccccc} (X \otimes Y) \otimes Z & \xrightarrow{a_{X,Y,Z}} & X \otimes (Y \otimes Z) & \xrightarrow{c_{X,Y \otimes Z}} & (Y \otimes Z) \otimes X \\ c_{X,Y} \otimes \text{id}_Z \downarrow & & & & \downarrow a_{Y,Z,X} \\ (Y \otimes X) \otimes Z & \xrightarrow{a_{Y,X,Z}} & Y \otimes (X \otimes Z) & \xrightarrow{\text{id}_Y \otimes c_{X,Z}} & Y \otimes (Z \otimes X) \end{array}$$

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$$\begin{array}{ccccc} X \otimes (Y \otimes Z) & \xrightarrow{a_{X,Y,Z}^{-1}} & (X \otimes Y) \otimes Z & \xrightarrow{c_{X \otimes Y,Z}} & Z \otimes (X \otimes Y) \\ \text{id}_X \otimes c_{Y,Z} \downarrow & & & & \uparrow a_{Z,X,Y} \\ X \otimes (Z \otimes Y) & \xrightarrow{a_{X,Z,Y}^{-1}} & (X \otimes Z) \otimes Y & \xrightarrow{c_{X,Z} \otimes \text{id}_Y} & (Z \otimes X) \otimes Y \end{array}$$

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Symmetric monoidal category

A braided monoidal category is **symmetric** if the braiding is involutive:

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In $\text{Rep}_{\mathbb{k}}(G)$, c_{VW} is the flip

$$v \otimes w \mapsto w \otimes v.$$

Rigidity: dual data

Let $\mathcal{C}$ be monoidal. A **left dual** of $X \in \mathcal{C}$ is an object X^* with morphisms

$$\mathrm{ev}_X : X^* \otimes X \rightarrow \mathbf{1}, \quad \mathrm{coev}_X : \mathbf{1} \rightarrow X \otimes X^*,$$

satisfying the two triangular identities.

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A **right dual** is defined analogously, with evaluation $X \otimes {}^*X \rightarrow \mathbf{1}$ and coevaluation $\mathbf{1} \rightarrow {}^*X \otimes X$. A monoidal category is **rigid** if every object has both a left and a right dual.

Rigidity: triangular identities

The left-dual triangular identities are the commutativity of the following two diagrams:

$$\begin{array}{c} X \xrightarrow{\text{coev}_X \otimes \text{id}_X} (X \otimes X^*) \otimes X \xrightarrow{a_{X, X^*, X}} X \otimes (X^* \otimes X) \xrightarrow{\text{id}_X \otimes \text{ev}_X} X \\ \searrow \text{id}_X \swarrow \\ X^* \xrightarrow{\text{id}_{X^*} \otimes \text{coev}_X} X^* \otimes (X \otimes X^*) \xrightarrow{a_{X^*, X, X^*}^{-1}} (X^* \otimes X) \otimes X^* \xrightarrow{\text{ev}_X \otimes \text{id}_{X^*}} X^* \end{array}$$

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Note that if X^* or *X exists for a given X then it is unique up to a unique isomorphism, and that in a symmetric category we have a canonical isomorphism ${}^*X \cong X^*$, so there is only one dual.

Rigidity in $\text{Rep}_{\mathbb{k}}(G)$

For $V \in \text{Rep}_{\mathbb{k}}(G)$, set

$$V^* = \text{Hom}_{\mathbb{k}}(V, \mathbb{k}),$$

with contragredient G -action. Evaluation and coevaluation are

$$\text{ev} : V^* \otimes V \rightarrow \mathbb{k}, \quad \varphi \otimes v \mapsto \varphi(v),$$

$$\text{coev} : \mathbb{k} \rightarrow V \otimes V^*, \quad 1 \mapsto \sum_i v_i \otimes v_i^*,$$

for a basis $\{v_i\}$ and its dual basis $\{v_i^*\}$.

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for a basis $\{v_i\}$ and its dual basis $\{v_i^*\}$. These maps are G -equivariant and satisfy the triangular identities.

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- $\text{End}(\mathbf{1}) = \mathbb{k}$;

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A braided tensor category is called **symmetric** if its braiding is symmetric.

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It is a whole world where we can do algebra and representation theory!

Examples of symmetric tensor categories

For example, $\text{Rep}_{\mathbb{k}}(G)$ is a symmetric tensor category. The most basic case is $G = 1$, which gives the category of finite dimensional $\mathbb{k}$ -vector spaces, $\text{Vec}_{\mathbb{k}}$.

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An affine group scheme over $\mathbb{k}$ is a group object G in the category of affine schemes over $\mathbb{k}$. Concretely, it is an affine $\mathbb{k}$ -scheme G whose algebra of regular functions $\mathcal{O}(G)$ is endowed with a structure of a commutative Hopf algebra.

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A **rational representation** of an affine group scheme G is a finite-dimensional $\mathcal{O}(G)$ -comodule V , i.e., a vector space V with a **coaction**

$$\rho : V \rightarrow V \otimes \mathcal{O}(G)$$

which defines an action of the algebra $\mathcal{O}(G)^*$ on V . It is easy to see that rational representations of an affine group scheme G form a symmetric tensor category $\mathbf{Rep}(G)$.

Tensor functors

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Let $\mathcal{C}, \mathcal{D}$ be monoidal categories. A **monoidal functor** $F : \mathcal{C} \rightarrow \mathcal{D}$ consists of a functor and coherent isomorphisms

$$J_{X,Y} : F(X) \otimes F(Y) \xrightarrow{\sim} F(X \otimes Y), \quad \nu : \mathbf{1}_{\mathcal{D}} \xrightarrow{\sim} F(\mathbf{1}_{\mathcal{C}}),$$

compatible with the associativity and unit data. A **braided (symmetric) monoidal functor** is a monoidal functor between braided (symmetric) monoidal categories compatible with the braidings.

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If $\mathcal{C}, \mathcal{D}$ are tensor categories then a monoidal functor $F : \mathcal{C} \rightarrow \mathcal{D}$ is called a **tensor functor** if it is $\mathbb{k}$ -linear and faithful (hence exact). A **fiber functor** is a symmetric tensor functor to the target $\mathbf{Vec}_{\mathbb{k}}$.

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Thus we would like to explain why the previous examples **are equivalent** to $\mathbf{Rep}(G)$ for some affine group scheme G as symmetric tensor categories.

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This is a **fiber functor**: a symmetric tensor functor to $\text{Vec}_{\mathbb{k}}$.

Tensor automorphisms of a fiber functor

Let $\omega : \mathcal{C} \rightarrow \mathbf{Vec}_{\mathbb{k}}$ be a fiber functor. Define the affine group functor

$$\underline{\mathrm{Aut}}_{\otimes}(\omega) : R \mapsto \mathrm{Aut}^{\otimes}(\omega_R), \quad \omega_R(X) = \omega(X) \otimes_{\mathbb{k}} R.$$

Thus an R -point of $\underline{\mathrm{Aut}}_{\otimes}(\omega)$ is a family

$$\{g_X : \omega(X) \otimes R \xrightarrow{\sim} \omega(X) \otimes R\}_{X \in \mathcal{C}}$$

which is natural in X , preserves tensor products and the unit, and is invertible.

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For $\mathcal{C} = \text{Rep}(G)$ and $\omega = \omega_G$, evaluation of representations gives

$$G \cong \underline{\text{Aut}}_{\otimes}(\omega_G).$$

The Coend of matrix coefficients

The functor $\underline{\text{Aut}}_{\otimes}(\omega)$ is represented by a Hopf algebra constructed from matrix coefficients. Set

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Concretely, this is the quotient of

$$\bigoplus_{X \in \mathcal{C}} \omega(X)^* \otimes \omega(X)$$

by the relations

$$\omega(f)^* \varphi \otimes v - \varphi \otimes \omega(f)v \in \omega(Y)^* \otimes \omega(Y)$$

for all morphisms $f : X \rightarrow Y$, where $\varphi \in \omega(Y)^*$ and $v \in \omega(X)$.

Hopf algebra structure on the Coend

Write the class of $\psi \otimes v \in \omega(X)^* \otimes \omega(X)$ as $[\psi \mid v]_X$. The coalgebra structure is the usual one for matrix coefficients:

$$\Delta([\psi \mid v]_X) = \sum_i [\psi \mid e_i]_X \otimes [e_i^* \mid v]_X, \quad \varepsilon([\psi \mid v]_X) = \psi(v),$$

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The tensor structure of ω defines multiplication:

$$[\psi \mid v]_X [\varphi \mid w]_Y = [\psi \otimes \varphi \mid v \otimes w]_{X \otimes Y},$$

after identifying $\omega(X \otimes Y) \cong \omega(X) \otimes \omega(Y)$. Duals give the antipode.

Tannakian reconstruction

Let $\mathcal{C}$ be a symmetric tensor category and let

$$\omega : \mathcal{C} \rightarrow \mathbf{Vec}_{\mathbb{k}}$$

be a fiber functor. Put

$$G := \underline{\mathbf{Aut}}_{\otimes}(\omega), \quad \mathcal{O}(G) = \int^{X \in \mathcal{C}} \omega(X)^* \otimes \omega(X).$$

Then the canonical coactions on $\omega(X)$ define a symmetric tensor equivalence

$$\mathcal{C} \simeq \mathbf{Rep}(G).$$

This shows that all our examples are of the form $\mathbf{Rep}(G)$ for an affine group scheme G .

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It was shown by Deligne and Milne that the fiber functor is unique up to isomorphism; consequently G is uniquely determined up to conjugation.

Abstract groups and proalgebraic completion

Let us now explain how this works in our first example. Let Γ be an abstract group. The forgetful functor gives

$$\hat{\Gamma} := \underline{\mathrm{Aut}}_{\otimes}(\omega_{\Gamma}), \quad \mathrm{Rep}_{\mathbb{k}}(\Gamma) \simeq \mathrm{Rep}(\hat{\Gamma}).$$

The affine group scheme $\hat{\Gamma}$ is the **proalgebraic completion** of Γ over $\mathbb{k}$.

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Its coordinate algebra is the subalgebra of $\mathrm{Fun}(\Gamma, \mathbb{k})$ spanned by the matrix coefficients

$$g \mapsto \langle f, gv \rangle, \quad v \in V, \quad f \in V^*, \quad V \in \mathrm{Rep}_{\mathbb{k}}(\Gamma).$$

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Thus the group scheme $\widehat{\Gamma}$ is reduced, i.e., it is a proalgebraic group. Namely, it is the inverse limit of algebraic groups G equipped with a homomorphism from Γ , i.e.,

$$\widehat{\Gamma} = \varprojlim_{\Gamma \rightarrow G} G.$$

In particular, $\widehat{\Gamma}$ carries a canonical (universal) homomorphism $\Gamma \rightarrow \widehat{\Gamma}$ (which may or may not be injective).

Tannakian categories

A symmetric tensor category $\mathcal{C}$ over $\mathbb{k}$ is called **Tannakian** if it admits a fiber functor

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Thus Tannakian categories are precisely the symmetric tensor categories whose objects can be realized as finite-dimensional vector spaces, functorially and compatibly with tensor products, duals, and the symmetry.

So it is natural to ask: **are there other examples of symmetric tensor categories?**

The first non-Tannakian example - supervector spaces

The answer turns out to be **yes**. Assume $\text{char } \mathbb{k} \neq 2$. Let $\text{sVec}_{\mathbb{k}}$ be the category of finite-dimensional **supervector spaces**

$$V = V_{\bar{0}} \oplus V_{\bar{1}}.$$

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The underlying tensor category is equivalent to $\text{Rep}_{\mathbb{k}}(\mathbb{Z}/2)$, but the symmetric structure is different. But why is it a genuinely new example?

Trace and categorical dimension

Let X be an object of a symmetric tensor category and let $f : X \rightarrow X$. The **categorical trace** is

$$\mathrm{Tr}(f) : \mathbf{1} \xrightarrow{\mathrm{coev}_X} X \otimes X^* \xrightarrow{f \otimes 1} X \otimes X^* \xrightarrow{c_{X, X^*}} X^* \otimes X \xrightarrow{\mathrm{ev}_X} \mathbf{1}.$$

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In $\mathrm{Vec}_{\mathbb{k}}$, the trace is the usual trace and dimension is the usual dimension viewed as an element of $\mathbb{k}$. In $\mathrm{sVec}_{\mathbb{k}}$, the trace is the **supertrace** $\mathrm{Tr}(f) = \mathrm{Tr}(f_0) - \mathrm{Tr}(f_1)$, and the dimension is the **superdimension**

$$\dim(V) = \dim_{\mathbb{k}} V_{\bar{0}} - \dim_{\mathbb{k}} V_{\bar{1}}.$$

Invertible objects distinguish $\mathbf{sVec}$ from Tannakian categories

An object X in a monoidal category is **invertible** if there exists Y such that

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In $s\mathbf{Vec}_{\mathbb{k}}$, the one-dimensional odd line $\mathbb{k}^{0|1}$ is invertible and satisfies

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How can we generalize this example?

Affine supergroup schemes

Affine supergroup schemes

An **affine superscheme** X over $\mathbb{k}$ is represented by a commutative algebra $\mathcal{O}(X) = \mathcal{O}(X)_{\bar{0}} \oplus \mathcal{O}(X)_{\bar{1}}$ in $\mathbf{sVec}_{\mathbb{k}}$ (a.k.a. **supercommutative algebra**).

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Equivalently, G is a representable group-valued functor of points on supercommutative $\mathbb{k}$ -algebras.

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Equivalently, G is a representable group-valued functor of points on supercommutative $\mathbb{k}$ -algebras.

Basic examples of affine supergroup schemes are classical algebraic supergroups, such as $GL(m|n)$ and $Osp(m|2n)$.

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Equivalently, G is a representable group-valued functor of points on supercommutative $\mathbb{k}$ -algebras.

Basic examples of affine supergroup schemes are classical algebraic supergroups, such as $GL(m|n)$ and $Osp(m|2n)$.

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There is a slight generalization of this example. Suppose $z \in G(\mathbb{k})$ is an element of order 2 acting on $\mathcal{O}(G)$ by parity. Then we may consider the category $\mathcal{C} \simeq \mathbf{Rep}(G, z)$ of representations of G on which z acts by parity. This is also a symmetric tensor category, and $\mathbf{Rep}(G) \cong \mathbf{Rep}(\{1, z\} \ltimes G, z)$, so it generalizes the previous example.

Superfiber functors and super-Tannakian categories

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Super-Tannakian reconstruction

Let

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be a superfiber functor. Deligne showed that in this case $\mathcal{C} = \mathbf{Rep}(G, z)$ for a unique pair (G, z) up to conjugation.

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The coordinate Hopf superalgebra is again a Coend of matrix coefficients:

$$\mathcal{O}(G) = \int^{X \in \mathcal{C}} \omega(X)^* \otimes \omega(X),$$

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Since ω is unique, so is (G, z) .

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We have thus enlarged the Tannakian class from fiber functors to $\text{Vec}_{\mathbb{k}}$ to fiber functors to $\text{sVec}_{\mathbb{k}}$, i.e., to super-Tannakian categories. The next question is:

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However, the answer in characteristic zero becomes **yes** under the condition of **moderate growth**.

Moderate growth

Let $\mathcal{C}$ be a tensor category. For $X \in \mathcal{C}$, let

$$\ell_n(X) := \ell(X^{\otimes n})$$

be the length of the n -th tensor power. The category $\mathcal{C}$ has **moderate growth** if for every X there exists $C_X > 0$ such that

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Thus every super-Tannakian category has moderate growth.

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Let $\mathbb{k}$ be algebraically closed of characteristic 0. A symmetric tensor category over $\mathbb{k}$ has moderate growth if and only if it is super-Tannakian.

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Equivalently, under moderate growth there exists a superfiber functor

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This shows that while we can do algebra in any symmetric tensor category, in characteristic zero and moderate growth setting such algebra can be described in classical terms, as ordinary equivariant algebra under a supergroup scheme.

Lecture 2

Deligne categories without moderate growth

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The example is Deligne's interpolation $\text{Rep } GL_t$ of $\text{Rep}_{\mathbb{C}} GL_n$ in which n is replaced by a parameter $t \in \mathbb{C}$.

For $t \notin \mathbb{Z}$, the category $\text{Rep } GL_t$ is semisimple symmetric tensor category, but it is not super-Tannakian because it does not have moderate growth.

The classical input: mixed tensor powers

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Thus one studies the spaces

$$\text{Hom}_{GL_n}([r, s]_n, [p, q]_n)$$

and their behavior when n is large relative to r, s, p, q .

By rigidity in $\text{Rep } GL_n$, there is a canonical identification

$$\text{Hom}_{GL_n}([r, s]_n, [p, q]_n) \cong \text{Hom}_{GL_n}(V^{\otimes r} \otimes V^{\otimes q}, V^{\otimes p} \otimes V^{\otimes s}) = \text{Hom}_{GL_n}(V^{\otimes r+q}, V^{\otimes p+s}).$$

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This is clearly zero unless $r + q = s + p$. If this holds, then the first fundamental theorem of invariant theory for GL_n (i.e., Schur-Weyl duality) identifies these spaces, in the stable range $n \geq r + q = s + p$, with spaces spanned by contraction-permutation diagrams.

Stable Hom spaces

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The dependence on n is simple: every closed loop produced in a composition contributes the scalar n .

Walled Brauer diagrams: the wall rule

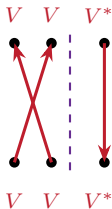
The stable Hom spaces are described by **walled Brauer diagrams**. A diagram for a morphism $[r, s] \rightarrow [p, q]$ has a vertical wall separating the contravariant side (r copies of V at the bottom and p at the top, left of the wall) from the covariant side (s copies of V^* at the bottom and q at the top, right of the wall).

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- A strand which does not cross the wall goes from the bottom row to the top row on the left and from the top row to the bottom row on the right.

no wall crossing



crossing: top-top



crossing: bottom-bottom

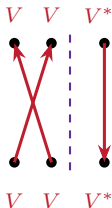


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- A strand which crosses the wall bends back: it connects two top endpoints (oriented to the left), or two bottom endpoints (oriented to the right).

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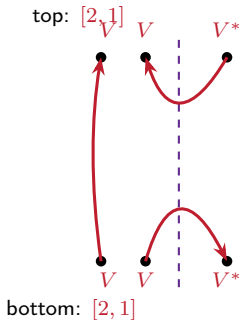
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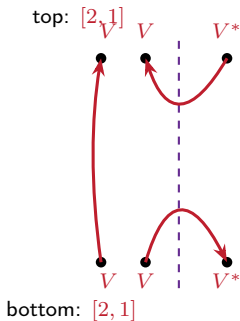
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The diagram encodes a matching together with the orientation rule; there is no difference between over-crossings and under-crossings.

Composition and the loop scalar

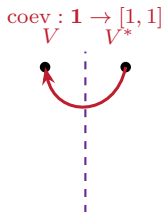
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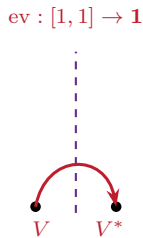
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The basic example is the composition

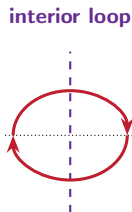
$$\mathbf{1} \xrightarrow{\text{coev}} [1, 1] \xrightarrow{\text{ev}} \mathbf{1}, \quad \text{ev} \circ \text{coev} = n \text{ id}_{\mathbf{1}}.$$



$\circ$



$=$



remove loop; multiply by n

Polynomiality and interpolation

For fixed p, q, r, s and large n , the composition gives bilinear maps

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This gives the diagrammatic interpolation of the tensor category generated by the defining representation of GL_n and its dual.

The category $\widetilde{\text{Rep}}GL_t$

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The tensor product is $[p, q] \otimes [r, s] = [p + r, q + s]$ (**putting diagrams next to each other and moving the V to left and V^* of the right**) and duality is $[r, s]^* = [s, r]$ (**rotating diagrams 180°**). The unit object is $\mathbf{1} = [0, 0]$. The braiding is swapping two diagrams which are next to each other.

Symmetric pseudotensor categories

Definition. A **symmetric pseudotensor category** over $\mathbb{k}$ is a Karoubian (i.e., additive idempotent complete) $\mathbb{k}$ -linear rigid symmetric monoidal category $\mathcal{C}$ such that

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Lemma. If $W_{r,s}(t)$ is semisimple for all r, s , then $\text{Rep} GL_t$ is a semisimple tensor category.

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Namely, $V_{\lambda,\mu}$ interpolates the irreducible GL_n -module with highest weight

$$(\lambda_1, \dots, \lambda_r, 0, \dots, 0, -\mu_s, \dots, -\mu_1).$$

for $n \geq r + s$.

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Thus

$$\ell(V^{\otimes d}) \geq \sqrt{d!},$$

which is faster than exponential growth.

Tensor ideals

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The **quotient** $\mathcal{C}/\mathcal{I}$ is the symmetric pseudotensor category which has the same objects as $\mathcal{C}$ and morphism spaces

$$\text{Hom}_{\mathcal{C}/\mathcal{I}}(X, Y) = \text{Hom}_{\mathcal{C}}(X, Y) / \mathcal{I}(X, Y).$$

Negligible morphisms

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The negligible morphisms form a tensor ideal, denoted $\mathcal{N} \subseteq \mathcal{C}$.

Semisimplicity of the negligible quotient

Lemma. (Benson) Suppose that for every object $X \in \mathcal{C}$, every nilpotent endomorphism $a \in \text{End}_{\mathcal{C}}(X)$ satisfies $\text{Tr}(a) = 0$ (**trace zero property**).

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(ii) is straightforward.

Corollary If $\mathcal{C}$ has trace zero property then the quotient $\mathcal{C}/\mathcal{N}$ is a semisimple symmetric tensor category, with simples being the indecomposables of $\mathcal{C}$ of nonzero dimension.

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Consequently,

$$(\operatorname{Rep} GL_t)^{ss} \simeq \operatorname{Rep}_{\mathbb{C}} GL_n.$$

Abelian envelopes and universal property

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If $t \notin \mathbb{Z}$, write $\text{Rep}^{ab} GL_t := \text{Rep } GL_t$. Then for all $t \in \mathbb{C}$ $\text{Rep } GL_t$ have the following **universal property**: symmetric tensor functors $H : \text{Rep}^{ab} GL_t \rightarrow \mathcal{D}$ to a symmetric tensor category $\mathcal{D}$ over $\mathbb{C}$ correspond to objects $X \in \mathcal{D}$ of non-moderate growth and dimension t . Namely, $X = H([1, 0])$.

Ultrafilters

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Alternatively, one may say that an ultrafilter on I is a homomorphism $\chi : \text{Fun}(I, \mathbb{F}_2) \rightarrow \mathbb{F}_2$, i.e., a maximal ideal ($\text{Ker}\chi$) in this ring. Then $A \in \mathcal{F}$ iff $\chi(1_A) = 1$.

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- if $A, B \in \mathcal{F}$, then $A \cap B \in \mathcal{F}$;
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Alternatively, one may say that an ultrafilter on I is a homomorphism $\chi : \text{Fun}(I, \mathbb{F}_2) \rightarrow \mathbb{F}_2$, i.e., a maximal ideal ($\text{Ker}\chi$) in this ring. Then $A \in \mathcal{F}$ iff $\chi(1_A) = 1$.

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Ultraproducts of sets and algebraic structures

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is the set of functions

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First-order structures and first-order properties are compatible with this construction. For example, ultraproducts of groups, rings, modules, and fields are again groups, rings, modules, and fields.

Ultraproducts of sets and algebraic structures

More precisely, a property P is said to hold **for $\mathcal{F}$ -almost all i** if

$$\{i \in I \mid X_i \text{ satisfies } P\} \in \mathcal{F}.$$

By Łoś's theorem, any first-order property which holds for $\mathcal{F}$ -almost all i holds for the ultraproduct

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Also K has cardinality continuum. Hence, by Steinitz's theorem,

$$K \cong \mathbb{C}.$$

Ultraproducts of sets and algebraic structures

As another example, let I be the set of primes and consider

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What is its characteristic? For every fixed prime ℓ , the characteristic is not ℓ , since $\text{char } \overline{\mathbb{F}}_p \neq \ell$ for $\mathcal{F}$ -almost all primes p (in fact, all but one). Hence

$$\text{char } L = 0.$$

Therefore, again by Steinitz's theorem,

$$L \cong \mathbb{C}.$$

Lecture 3

Categorical ultraproducts

Let $\mathcal{F}$ be a non-principal ultrafilter on $\mathbb{N}$, and let $\mathcal{C}_n$ be essentially small K_n -linear categories. One can form the **categorical ultraproduct**

$$\mathcal{U} := \prod_{n \in \mathbb{N}}^{\mathcal{F}} \mathcal{C}_n.$$

Its objects are sequences $X = (X_n)$, $X_n \in \mathcal{C}_n$ defined for n in some domain $A_X \in \mathcal{F}$, and its Hom spaces are

$$\mathrm{Hom}_{\mathcal{U}}(X, Y) = \prod_{n \in \mathbb{N}}^{\mathcal{F}} \mathrm{Hom}_{\mathcal{C}_n}(X_n, Y_n),$$

linear over the ultraproduct field $K := \prod_{n \in \mathbb{N}}^{\mathcal{F}} K_n$.

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If the $\mathcal{C}_n$ are rigid symmetric monoidal, then so is $\mathcal{U}$, with tensor product, duality, associativity, and symmetry defined componentwise.

However, $\mathcal{U}$ need not be a symmetric tensor category. The condition that every object have finite length is not first-order. Similarly, finite dimensionality of Hom spaces need not be preserved by the ultraproduct.

The generated subcategory has finite length

Now take

$$\mathcal{U} = \prod_{n \in \mathbb{N}}^{\mathcal{F}} \operatorname{Rep}_{\overline{\mathbb{Q}}} GL_n, \quad V = (\overline{\mathbb{Q}}, \overline{\mathbb{Q}}^2, \overline{\mathbb{Q}}^3, \dots).$$

Although $\mathcal{U}$ itself may fail the artinian condition, we restrict to the full tensor subcategory

$$\langle V \rangle \subset \mathcal{U}$$

generated by V : its objects are subquotients of finite direct sums of

$$V^{\otimes r} \otimes V^{*\otimes s}, \quad r, s \geq 0.$$

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Hence their dimensions stabilize. Consequently the Hom spaces in $\langle V \rangle$ are finite-dimensional, and all objects of $\langle V \rangle$ have finite length. Thus $\langle V \rangle$ is a symmetric tensor category.

The dimension parameter in the ultraproduct

The categorical dimension of V is the ultraproduct of the ordinary dimensions:

$$\dim(V) = [1, 2, 3, \dots] \in K = \prod_{n \in \mathbb{N}}^{\mathcal{F}} \overline{\mathbb{Q}} \cong \mathbb{C}.$$

Choose a field isomorphism

$$\xi : K \xrightarrow{\sim} \mathbb{C}.$$

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$$t := \xi([1, 2, 3, \dots]) \in \mathbb{C}$$

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Deligne's theorem states that

$$\langle V \rangle \simeq \operatorname{Rep} GL_t$$

for this transcendental value of t . Indeed, by the universal property of $\operatorname{Rep} GL_t$, there is a unique symmetric tensor functor $H : \operatorname{Rep} GL_t \rightarrow \langle V \rangle$ with $H([1, 0]) = V$, and it is easy to check that this functor is an equivalence.

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If $t' \in \mathbb{C}$ is another transcendental number, an isomorphism

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Thus, for a fixed non-principal ultrafilter, the same categorical ultraproduct realizes

$$\text{Rep } GL_t$$

for every transcendental t , after a suitable identification of the ultraproduct field with $\mathbb{C}$.

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Proof. For large N , there are roughly $N^{1/\deg(q)}$ positive integers of the form $q(n)$ with $q(n) < N$, whereas for a fixed collection of m primes there are on the order of $(\log N)^m$ positive integers less than N divisible only by those primes.

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Recall that the field ultraproduct $K = \prod_{p \in \mathbb{A}_q}^{\mathcal{F}} \overline{\mathbb{F}}_p$ may be identified with $\mathbb{C}$. Under a suitable identification, the element $[n_p] \in K$ maps to the prescribed root t of q .

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With $V = (\overline{\mathbb{F}}_p^{n_p})_p$, from the universal property of $\text{Rep } GL_t$ one obtains

$$\text{Rep } GL_t \simeq \langle V \rangle \subset \prod_{p \in \mathbb{A}_q}^{\mathcal{F}} \text{Rep}_{\overline{\mathbb{F}}_p}(GL_{n_p}).$$

Integral parameters and abelian envelopes

The case $t \in \mathbb{Z}$ (so $q(x) = x - t$, $\overline{n_p} = t$) is the same, except $n_p \rightarrow \infty$ is not automatic and needs to be arranged. For example, one may take $n_p = p + t$.

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Fixed positive characteristic

One may also use the ultrafilter approach to construct symmetric tensor categories of non-moderate growth in positive characteristic.

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One may also use the ultrafilter approach to construct symmetric tensor categories of non-moderate growth in positive characteristic. To this end, fix a prime p and work over algebraically closed fields of characteristic p . Consider

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But now we encounter a new phenomenon. We have $\dim(V) = [1, 2, 3, \dots] \in K = \prod_n^{\mathcal{F}} \overline{\mathbb{F}}_p$. However, this is not an arbitrary element.

Why dimensions lie in $\mathbb{F}_p$

Lemma. Let $\mathcal{C}$ be a symmetric tensor category over a field of characteristic $p > 0$, and let $X \in \mathcal{C}$ have categorical dimension d . Then $d \in \mathbb{F}_p$.

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Exterior powers and p -adic dimension

The missing information is contained in the dimensions of exterior powers. For $X \in \mathcal{C}$, set

$$t_i := \dim(\wedge^{p^i} X) \in \mathbb{F}_p, \quad i \geq 0.$$

These digits define the p -adic dimension

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Equivalently, one can show that $\mathrm{Dim}(X)$ is characterized by the generating function

$$\sum_{m \geq 0} \dim(\wedge^m X) z^m = (1 + z)^{\mathrm{Dim}(X)},$$

where, for $t = \cdots t_2 t_1 t_0$,

$$(1 + z)^t = (1 + z)^{t_0} (1 + z^p)^{t_1} (1 + z^{p^2})^{t_2} \cdots.$$

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It turns out that $t = \lim_{n \rightarrow \infty}^{\mathcal{F}} n$. Moreover, a good news is that the resulting category $\langle V \rangle$ depends only on t , not on the remaining nonconstructive choices. We denote this category by

$$\mathrm{Rep}^{ab} GL_t, \quad t \in \mathbb{Z}_p.$$

Thus in characteristic p , interpolation of rank is naturally p -adic.

From interpolation to moderate growth

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The category $\text{Rep}_{\mathbb{k}}(\mathbb{Z}/p)$

Let $\text{char } \mathbb{k} = p$. Then

$$\mathbb{k}[\mathbb{Z}/p] \cong \mathbb{k}[g]/(g^p - 1) = \mathbb{k}[g]/(g - 1)^p.$$

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Equivalently, $J_n \cong \mathbb{k}^n$, and g acts by one unipotent Jordan block of size n .

The categorical dimension of J_n is $n \in \mathbb{k}$. Hence

$$\dim(J_p) = p = 0.$$

By the criterion from Lecture 2, the dimension-zero indecomposable J_p becomes zero after semisimplification, while $J_1, \dots, J_{p-1}$ give the simple objects.

Definition of Ver_p

The **Verlinde category** is the semisimplification

$$\text{Ver}_p := \text{Rep}_{\mathbb{k}}(\mathbb{Z}/p)^{ss}.$$

Let

$$S : \text{Rep}_{\mathbb{k}}(\mathbb{Z}/p) \rightarrow \text{Ver}_p$$

be the quotient functor. The simple objects of Ver_p are

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The category $\mathbf{Ver}_p$ is a **symmetric fusion category**: it is semisimple, rigid, symmetric monoidal, and has finitely many simple objects.

It is the basic positive-characteristic replacement for $\mathbf{sVec}$.

The Verlinde fusion rule

The tensor product in Ver_p is the truncated Clebsch–Gordan rule:

$$L_m \otimes L_n = \bigoplus_{i=1}^{\min(m,n,p-m,p-n)} L_{|m-n|+2i-1}.$$

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The same fusion rules occur for $\widehat{\mathfrak{sl}}_2$ at level $p-2$ (**Verlinde algebra** in the **2-dimensional WZW conformal field theory**), which explains the name “Verlinde category”.

Small primes and the first exotic example

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If a fiber functor $\text{Ver}_5 \rightarrow \text{sVec}_{\mathbb{k}}$ existed, the integer superdimension d of the image of L_3 would satisfy

$$d^2 = 1 + d.$$

This equation has no integral solution. Hence Ver_5 is not super-Tannakian.

Moderate growth of Ver_p

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More generally, a similar argument shows that for all $p \geq 5$, Ver_p is a symmetric tensor category of moderate growth which is not super-Tannakian. This is the simplest way in which Deligne's characteristic-zero theorem fails in characteristic p .

The copy of $s\mathbf{Vec}$ inside $\mathbf{Ver}_p$

For $p > 2$, the last simple object satisfies

$$L_{p-1} \otimes L_{p-1} \cong L_1, \quad \dim(L_{p-1}) = p - 1 = -1.$$

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Let $\mathbf{Ver}_p^+$ be the tensor subcategory generated by the odd-indexed simple objects:

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Then, for $p > 2$,

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Thus $\mathbf{Ver}_p$ contains $\mathbf{sVec}_{\mathbb{k}}$, but it is strictly larger when $p \geq 5$.

Other constructions of $\mathbf{Ver}_p$

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These realizations connect Ver_p with modular Lie theory, tilting modules, and quantum groups.